

MOD 305 Course Outline

MOD 305 — Recursive Dynamic CGE

Course Outline

School	03 Economic Modelling
Level	300 (practitioner / intermediate to advanced)
Hours	45 (indicative band 40–50)
Shape	5 parts · 13 sessions · 8 laboratories · 4 assignments · 1 examination · 1 final project
Model and data	CDCGE - AICTEM Country-Disaggregated CGE (ENVISAGE 10.01 structure) · GTAP 11 (v11b), reference year 2017, values in USD million
Delivery	Instructor-led — online, blended or intensive in person
Prerequisites	MOD 303. MOD 304 is strongly recommended: this course assumes the learner already reads a parameter file before a model and a termination status before a variable. Familiarity with difference equations and present-value arithmetic.
Assessment	Pass 60 %, with 60 % laboratory competency
Version	1.0 · reviewed 2026-08-27 · 12-month cycle

What this course is

A recursive dynamic CGE model solves a sequence of static equilibria, carrying capital and productivity forward between them. This course builds that sequence: the accumulation equation, the exogenous drivers, the baseline, the counterfactual path, and the discounting that turns a path into a welfare number. It then reads the twelve-period path this school's own model produced — and finds that four of its periods did not solve, that the path continues through them, and that on the factor side the model is not accumulating at all. Both findings are invisible in the series and obvious in the status column.

The course is built on one real recursive-dynamic run and is candid about what that run can support. Four of its twelve periods did not solve, so most of the path may not be quoted at all — school gap M-14 — and the capital stock does not accumulate into the factor market, so no investment-to-output statement can be made from it — school gap M-15. Both were found by two tests the course teaches and then applies to its own material. All 51 registered findings are in the appendix.

The rule the course is built on

A smooth path is not evidence that it solved. A plotted trajectory interpolates through periods that terminated badly and looks exactly like one that did not, which is why the status column is read before the chart is drawn. The second habit is the same one applied to the baseline: a projection is conditional on drivers the model does not forecast, and reporting it without them is reporting somebody else's assumption as your result.

Who it is for

Modellers moving from comparative statics to a path, and analysts who must read a published projection and say what part of it is a model result and what part is a driver assumption carried forward.

Structure

Part	Title	Sessions	Laboratory	Hours
P1	Time, accumulation and the exogenous drivers	01–03	01	7.5
P2	Building a baseline and reading a path	04–05	02–03	7.5
P3	Does the capital stock accumulate?	06–07	04	6.0
P4	Counterfactuals, contamina- tion and welfare	08–10	05–06	9.0
P5	Emissions, sensitivity and reporting a projection	11–13	07–08	9.0
	Assignment and assessment time			6.0
	Total	13	8	45.0

The 13 sessions and 8 laboratories, in teaching order, and the subject matter of each:

P1 Time, accumulation and the exogenous drivers — Sessions 01–03 · 7.5 h · Laboratory 01

- **01 Time in a CGE Model** — Comparative static: two states of one year; Recursive dynamic: a sequence of static equilibria; Intertemporal: all periods solved together; What each family can and cannot answer; Why a path is not a forecast; Where this school's model sits
- **02 The Accumulation Equation** — $K' = (1 - \delta)K + I$, and what each term assumes; Depreciation as a parameter, not an observation; Where investment comes from: the closure, again;

Putty-clay against putty-putty capital; Vintage structure and why most models omit it; How to test whether a model's capital actually accumulates

- **03 Exogenous Drivers** — Labour force growth; Total factor productivity growth; Autonomous energy efficiency improvement; World prices and world demand; Policy already legislated; The driver list as the real content of a baseline
- *Lab 01 The Driver Table* — What does this baseline assume, and where did each assumption come from?

P2 Building a baseline and reading a path — Sessions 04–05 · 7.5 h · Laboratory 02–03

- **04 Building a Baseline** — What a baseline is for; Calibrating drivers to a target growth rate; The difference between a baseline and a forecast; Business-as-usual, and what it assumes about policy; Reporting a baseline honestly; The baseline this extraction carries
- **05 Reading a Path** — The per-period termination status; Why a failed period contaminates its successors; Reading growth rates rather than levels; Identities that must hold along a path; Discontinuities, and what their absence does not prove; The status column of this school's own path
- *Lab 02 Read the Path* — What does this baseline path say?
- *Lab 03 The Status Column* — Did every period solve?

P3 Does the capital stock accumulate? — Sessions 06–07 · 6.0 h · Laboratory 04

- **06 Does the Capital Stock Accumulate?** — The test: is employment a geometric trend?; The test: does $K' = (1 - \delta)K + I$ hold?; What an exogenous factor path means for a result; Capital supply against capital employed; What this school's model does; Why the distinction is not academic
- **07 What a Genuine Accumulation Block Requires** — The accumulation identity, wired to the factor market; An investment allocation rule across sectors; A depreciation rate, by asset type if possible; Adjustment costs, and whether to include them; The rate of return that closes the investment decision; What each addition costs in assumptions
- *Lab 04 Does the Capital Stock Accumulate?* — Is this model actually dynamic on the factor side?

P4 Counterfactuals, contamination and welfare — Sessions 08–10 · 9.0 h · Laboratory 05–06

- **08 Counterfactual Paths** — A policy path against a baseline path; Permanent, temporary and phased shocks; Announcement effects, and why this family has none; Reading a difference rather than a level; Path dependence and hysteresis; What this extraction does not carry
- **09 Contamination** — Initialisation from the previous period's state; Why the sequence does not self-correct; Why the series looks smooth anyway; What may be reported from a contaminated path; Restart strategies, and their cost; This school's own path, worked through
- **10 Welfare Over a Path** — Present value and the discount rate; Pure time preference against growth discounting; The Ramsey formula and its parameters; Why the discount rate usually dominates the result; Terminal conditions and the truncation problem; Reporting a discounted welfare number
- *Lab 05 Growth Accounting Along the Path* — What is driving the growth in this baseline?
- *Lab 06 Discounted Welfare* — What is this path worth, and at what discount rate?

P5 Emissions, sensitivity and reporting a projection — Sessions 11–13 · 9.0 h · Laboratory 07–08

- **11 Emissions and Climate Blocks** — Emissions as an output of the economic model; The autonomous energy efficiency improvement; Concentration, forcing and temperature; Damage functions, and their calibration; Stylised model units against an inventory; The emissions path

this extraction produces

- **12 Dynamic Sensitivity Analysis** — The driver assumptions as the first thing to sweep; The discount rate as the second; Why a dynamic sweep costs more than a static one; The convergence problem, compounded over periods; Reporting a range of paths; What this school did not run
- **13 Reporting a Projection** — What a modelled path is conditional on; Dates, and when they are legitimate; The driver table as a required annex; Reporting the status of every period; The difference between a projection and a forecast; The projection note template
- *Lab 07 The Emissions Path* — What happens to emissions in this baseline?
- *Lab 08 The Projection Note* — What can honestly be said about this path?

What a graduate can do

1. Distinguish comparative-static, recursive dynamic and intertemporal models, and say what each can answer.
2. Write the capital accumulation identity and state what it requires.
3. List every exogenous driver a dynamic baseline needs and say where each comes from.
4. Construct a baseline path and explain why it is an assumption rather than a forecast.
5. Read a per-period termination status before reading any series.
6. Explain how a single failed period contaminates every period after it.
7. Establish from the data whether a model's capital stock actually accumulates.
8. Compute and interpret a present-value welfare measure over a path.
9. Decompose a path result into baseline drivers and policy effect.
10. Design a dynamic sensitivity analysis, and say why it is harder than a static one.
11. Report a projection with the conditionality it actually carries.

The formal learning outcomes, and the map from each to the assessment that tests it, are in 12_LMS/learning_outcomes.md.

Assessment

Component	Weight	What it is
Knowledge quizzes	—	A bank of 117 items, used for formative practice and drawn on by the theory paper, which carries the marks
Laboratory exercises	35.04 %	8 laboratories, equally weighted. A laboratory is not complete until its validation checks have been read and its named deliverable produced: an irreproducible result is no result
Applied assignments	20 %	The driver table; The path diagnosis; Growth accounting and valuation; The projection note

Component	Weight	What it is
Theory examination	20 %	Closed-book, drawn from the quiz bank and the review questions
Final project	25 %	A complete diagnosis of a modelled dynamic path — this one, or one the learner is responsible for — delivered as a nine-section projection note plus the diagnostic scripts that produced it

The per-laboratory weights are an equal split of the laboratory component rounded to two decimals, so the column sums to 100.04 rather than 100. The component weights are the binding ones.

Pass 60 % overall and 60 % laboratory competency. Merit 70 %. Distinction 80 %. One resubmission of a single failed component within eight weeks. A laboratory is not complete until every validation check has been read and the provenance of every figure recorded: an irreproducible result is no result.

Data and access

No restricted data and no household record. The extraction carries one baseline path and no counterfactual, and the carbon price is exactly zero in every period, so the course reports no policy result and no forecast. Where a chapter needs a counterfactual to teach a method, it says that the path shown is the specification and not a run.

Materials and set-up

A 121-page Technical Manual, 14 lecture decks of 147 slides, 8 laboratory decks of 101 slides, a Laboratory Manual, Instructor Guide and Learner Workbook, a 117-item quiz bank, 5 assessment files, 18 datasets, 4 code exercises, 3 case studies and the LMS package. The README lists every file, and COURSE_BUILD_REPORT.md records how each was produced.

CDCGE at D:/EDIP/models/CDCGE and Julia with `-pkgimages=no`. The recursive driver is slower than the static solve: allow the laboratory session the time the Instructor Guide budgets rather than the time the static courses needed.

The 51 findings registered against this school, and the bearing each has on this course, are carried in the Technical Manual's gap appendix and listed in `COURSE_VERSION.json`.

MOD 305 Recursive Dynamic CGE · version 1.0 · AICTEM EDIS Academy, School 03 Economic Modelling.